

Machine Learning Problem Types and Solutions

The following examples explain different machine learning problems in simple language suitable for school and college students.

1. Predicting House Prices

Scenario: Predicting House Prices

Problem Type: Regression

Why this problem type? House prices are numerical values that can vary continuously, so regression is used.

Step-by-step Process:

- Collect Data – Gather details of houses that were sold earlier, including size, bedrooms, location, and selling price.
- Prepare the Data – Fix missing information and convert location names into a format the computer can understand.
- Split the Dataset – Divide the data into training and testing sets.
- Choose a Suitable Algorithm – Select Linear Regression or Decision Tree Regression.
- Train the Model – Use the training data so the model can find patterns between house features and price.
- Evaluate the Results – Check accuracy using RMSE and R^2 Score.
- Predict House Prices – Estimate the price of new houses using the trained model.

2. Identifying Fraudulent Transactions

Scenario: Identifying Fraudulent Transactions

Problem Type: Classification

Why this problem type? A transaction can only belong to categories such as Fraudulent or Not Fraudulent, making it a classification problem.

Step-by-step Process:

- Collect Data – Gather historical transaction records labeled as fraudulent or genuine.
- Prepare the Data – Clean the dataset and convert categorical information into numerical values.
- Create Useful Features – Add information such as spending habits and transaction frequency.
- Split the Dataset – Create training and testing datasets.
- Choose a Suitable Algorithm – Use Logistic Regression, Random Forest, or Neural Networks.

- Train the Model – Help the model learn the difference between fraudulent and normal transactions.
- Evaluate the Results – Measure Accuracy, Precision, Recall, and F1-Score.
- Deploy the System – Use the model to detect suspicious transactions in real time.

3. Grouping Customers Based on Spending Habits

Scenario: Grouping Customers Based on Spending Habits

Problem Type: Clustering

Why this problem type? Customers are not already labeled into groups, so clustering helps find similar groups automatically.

Step-by-step Process:

- Collect Data – Gather purchase history, spending amount, and shopping frequency.
- Prepare the Data – Normalize values so all features are treated equally.
- Choose a Clustering Algorithm – Use K-Means, DBSCAN, or Hierarchical Clustering.
- Find the Best Number of Groups – Apply the Elbow Method.
- Run the Model – Group similar customers together.
- Analyze the Groups – Identify high, medium, and low-spending customers.
- Use the Results – Create targeted marketing campaigns for each group.

4. Predicting Employee Salaries

Scenario: Predicting Employee Salaries

Problem Type: Regression

Why this problem type? Salary is a numerical value, so regression is the most suitable approach.

Step-by-step Process:

- Collect Data – Gather employee details such as experience, education, job title, and salary.
- Prepare the Data – Handle missing values and convert job titles into numbers.
- Split the Dataset – Divide the dataset into training and testing sets.
- Choose a Suitable Algorithm – Use Linear Regression or Random Forest Regression.
- Train the Model – Learn how different factors influence salary.
- Evaluate the Results – Measure accuracy using MAE and R^2 Score.
- Predict Salaries – Estimate salaries for new employees.

5. Detecting Spam Emails

Scenario: Detecting Spam Emails

Problem Type: Classification

Why this problem type? Emails are classified into categories such as Spam or Not Spam.

Step-by-step Process:

- Collect Data – Gather labeled spam and non-spam emails.
- Prepare the Data – Convert email text into numerical values using TF-IDF or embeddings.
- Split the Dataset – Create training and testing sets.
- Choose a Suitable Algorithm – Use Naive Bayes, SVM, or Neural Networks.
- Train the Model – Learn common patterns found in spam emails.
- Evaluate the Results – Check Precision, Recall, and F1-Score.
- Deploy the Model – Automatically filter spam messages.

6. Customer Reviews Sentiment Analysis

Scenario: Customer Reviews Sentiment Analysis

Problem Type: Classification

Why this problem type? Reviews are categorized as Positive or Negative, so classification is used.

Step-by-step Process:

- Collect Data – Gather customer reviews with sentiment labels.
- Clean the Text Data – Remove unnecessary words and symbols.
- Convert Text into Features – Use TF-IDF or Word2Vec.
- Split the Dataset – Create training and testing datasets.
- Choose a Suitable Algorithm – Use Logistic Regression, Naive Bayes, or BERT.
- Train the Model – Learn patterns linked to positive and negative opinions.
- Evaluate the Results – Measure accuracy and F1-Score.
- Predict Sentiment – Classify new reviews as positive or negative.

7. Predicting Car Insurance Claims

Scenario: Predicting Car Insurance Claims

Problem Type: Classification

Why this problem type? The result is either Claim or No Claim, which makes it a classification problem.

Step-by-step Process:

- Collect Data – Gather claim history, driving behavior, and customer information.
- Prepare the Data – Handle missing values and encode categories.
- Split the Dataset – Divide data into training and testing sets.
- Choose a Suitable Algorithm – Use Logistic Regression, Decision Trees, or Neural Networks.

- Train the Model – Learn patterns from previous insurance records.
- Evaluate the Results – Measure Precision-Recall and AUC-ROC Score.
- Deploy the Model – Predict claim likelihood for new customers.

8. Recommending Movies Based on Viewing History

Scenario: Recommending Movies Based on Viewing History

Problem Type: Clustering

Why this problem type? Users are grouped by similar interests, so clustering is used.

Step-by-step Process:

- Collect Data – Gather viewing history, ratings, and preferred genres.
- Prepare the Data – Convert categories into numerical values.
- Choose a Clustering Algorithm – Use K-Means or Hierarchical Clustering.
- Find the Best Number of Groups – Use the Elbow Method.
- Run the Model – Group users with similar preferences.
- Analyze the Groups – Identify categories such as Action Lovers or Drama Fans.
- Recommend Movies – Suggest content based on group interests.

9. Predicting Patient Recovery Time

Scenario: Predicting Patient Recovery Time

Problem Type: Regression

Why this problem type? Recovery time is measured in days or weeks, which is a continuous numerical value.

Step-by-step Process:

- Collect Data – Gather patient age, medical history, lifestyle details, and recovery records.
- Prepare the Data – Handle missing information and normalize measurements.
- Choose a Suitable Algorithm – Use Linear Regression or Random Forest Regression.
- Train the Model – Learn relationships between patient factors and recovery duration.
- Evaluate the Results – Measure prediction quality using RMSE.
- Predict Recovery Time – Estimate recovery duration for new patients.

10. Predicting Student Exam Scores

Scenario: Predicting Student Exam Scores

Problem Type: Regression

Why this problem type? Exam scores are numerical values, so regression is the correct choice.

Step-by-step Process:

- Collect Data – Gather study hours, attendance, previous grades, and exam scores.
- Prepare the Data – Handle missing values and standardize numerical features.
- Split the Dataset – Create training and testing datasets.
- Choose a Suitable Algorithm – Use Linear Regression or Support Vector Regression.
- Train the Model – Learn how study habits affect exam performance.
- Evaluate the Results – Measure performance using RMSE and R^2 Score.
- Predict Exam Scores – Estimate scores for future students.